

Problem 11.14

On July 1, 1999 a person invested 1,000 in a fund for which the force of interest at time t is given by $\delta_t = 0.02(3 + 2t)$, where t is the number of years since January 1, 1999. Determine the accumulated value of the investment on January 1, 2000.

Solution

Time is measured from January 1, 1999, so the deposit on July 1, 1999 is made at $t = 0.5$. The accumulated value is required on January 1, 2000, which corresponds to $t = 1$.

For a time-varying force of interest, the accumulation factor from time t_1 to time t_2 is

$$\exp \left(\int_{t_1}^{t_2} \delta_t dt \right).$$

Therefore, the accumulated value of the deposit is

$$AV = 1000 \exp \left(\int_{0.5}^1 0.02(3 + 2t) dt \right).$$

Evaluate the integral:

$$\int_{0.5}^1 0.02(3 + 2t) dt = [0.06t + 0.02t^2]_{0.5}^1.$$

Substituting the limits gives

$$(0.06 + 0.02) - (0.06(0.5) + 0.02(0.5)^2) = 0.08 - 0.035 = 0.045.$$

Thus,

$$AV = 1000e^{0.045} \approx 1046.027860.$$

Hence, the accumulated value on January 1, 2000 is

$$\boxed{AV \approx 1046.03}.$$